

costs. Efficiency of the module is up to 90% and MTBF is 4 million hours.

Flex Power Modules

www.flexpowermodules.com

Industrial robot reference design

The open source, fully integrated TMCM-1617-GRIP-REF reference design integrates hardware based field-oriented control and three communication ports that shrink the design size of electronic robotic grippers and simplifies the development of industrial robotic end-of-arm tooling (EoAT). Designed to fit within the standard form-factor used for EoAT grippers, the TMCM-1617-GRIP-REF reference design supports industrial EtherCAT, IO-Link or RS-485 communication while providing software-programmable analogue and digital input/outputs. Advantages of this reference design include increased speed of time to market, reduced size, and increased productivity.

TRINAMIC Motion Control

www.trinamic.com

Wireless connectivity module

The ESP32-WROOM-DA achieves high-quality, wireless connectivity across large areas, using two complementary PCB antennas in different directions. A co-existence of Wi-Fi and Bluetooth (Classic & Low-Energy) connectivity with low-power consumption allows the module to bring to life a wide range of complex IoT application scenarios. The module has a built-in ESP32-D0WD-V3 chip, which has an Xtensa 32-bit LX6 dual-core processor with a maximum frequency of 240 MHz, along with 520kB SRAM, and 448kB ROM. While being used, the module works according to signal strength, selecting the strongest signal to ensure continuous communication. When a weak signal is detected, the module continues to work with an API call that helps it switch to the other antenna with a stronger signal.

Espressif Systems

www.espressif.com

5G New Radio modules

Based on the Qualcomm 315 5G IoT Modem-RF system, two new 5G New Radio (NR) modules, the RG500S and RM500S, support customers in building dedicated 5G devices for a variety of verticals including industrial IoT, retail, smart energy, and private 5G networks. They both enable 5G NR sub-6GHz bands in stand-alone mode offering backward compatibility with LTE networks. Embedded with rich interfaces, they incorporate high-speed USB 3.0/3.1, PCIe 3.0, U(SIM), RGMII, and more, making them suitable for diversified industrial and consumer 5G applications such as industrial routers, robots, automation, intelligent manufacturing, smart cities, energy distribution, precision agriculture, construction, and mining.

Quectel Wireless Solutions

www.quectel.com

Computer-on-module for vision apps

Extending the i.MX 8 ecosystem with a new starter set for AI accelerated intelligent embedded vision applications, the SMARC computer-on-module with i.MX 8M Plus processor utilises an integrated NXP neural processing unit. Delivering up to 2.3 TOPS of performance for deep learning based AI, it can run inference engines and libraries such as Arm Neural Network and TensorFlow Lite. It integrates seamlessly with Basler embedded vision software to give OEMs an application-ready solution platform for the development of next-generation AI accelerated embedded vision systems. Typical applications are in automated checkout terminals in retail to building safety, in-vehicle vision for navigation to surveillance systems. Industrial use cases include HMIs with vision based user identification and gesture based machine operation as well as vision supported robotics and industrial quality inspection systems.

congatec

www.congatec.com

Switching power supplies

To meet the market's demand, the LMF75-23Bxx is a new enclosed switching power

supply series with active PFC in the 305RAC family, which can be widely used in security, intelligent buildings, industrial control, communications, and other industrial fields. The active PFC LMF75-23Bxx series has a PF up to 0.98, features a compact size,

wide input voltage range of 85-305V AC, and accepts AC or DC input. The series also meets IEC/EN61000-4, CISPR32/EN55032, IEC/UL/EN62368, and GB4943 standards. Protections of output short-circuit, over-current, over-voltage, and over-temperature are included.

Mornsun

www.mornsun-power.com

Universal battery charger

This universal battery charger, made in India, can charge LiFePO4, Li-ion, and lead-acid batteries. One has to just select the type of battery and let this

amazing device do its magic. It is a microprocessor based product that makes use

of a high-frequency design to keep product size compact. Salient features include wide AC input voltage range, active power factor correction, battery protection, and rugged product design. The battery charger offers value to users in terms of portability, charging quality, accuracy, and product reliability.

Stellar Innovations

www.stellar.co.in

T&M SOLUTIONS

Portable RF spectrum analyser

5GPro spectrum analyser addresses the need of mobile network operators who need to support and maintain new 5G deployments as well as existing 4G/LTE installations. It is a modular, field-focused radio frequency (RF) testing solution that

provides accurate visibility into 4G/LTE and 5G NR environments, with the ability to analyse FR1(sub-6GHz) and FR2 (mmWave) bands using a single, field-upgradeable solution. It delivers accurate RF over-the-air measurements,